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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

DoH-Shield

Project Report Submitted by

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CERTIFICATE

Certified that the project work titled **'DoH-Shield'** is carried out by **Tejasvi Vasant Hegde (1RV23CS272) , Tarun R (1RV23CS271) , Tanmay Dev D (1RV23CS269)** , who are bonafide students of R. V. College of Engineering, Bengaluru, in partial fulfillment of the curriculum requirement of 5th Semester **Network Programming and Security(CS362IA)** Laboratory Mini Project during the academic year **2025-2026**. It is certified that all corrections/suggestions indicated for the internal Assessment have been incorporated in the report. The report has been approved as it satisfies the academic requirements in all respect laboratory mini-project work prescribed by the institution.

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External Examination

Name of Examiners	Signature with date
1	
2	

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Abstract

DNS-over-HTTPS (DoH, RFC 8484) encrypts DNS transactions inside HTTPS tunnels, preventing passive eavesdroppers from reading domain names in plaintext. However, encryption of packet payloads alone is insufficient to protect user privacy. Modern **Website Fingerprinting (WF)** attacks exploit observable traffic metadata—packet size distributions, flow durations, inter-packet arrival times, and request-response latencies—to identify visited websites with greater than 99% accuracy, even on encrypted DoH streams.

This report presents **DoH-Shield**, a mathematically rigorous, client-side traffic morphing proxy designed to neutralize state-of-the-art WF attacks. DoH-Shield operates as a local Man-in-the-Middle (MITM) proxy intercepting browser DoH queries on port 8080, extracting a 29-dimensional real-time statistical feature vector from each DNS burst session, mapping those features into pre-trained K-Means clusters representing indistinguishable traffic demographics, and morphing packet characteristics toward cluster centroids. Dynamic injection of EDNS(0)-padded dummy DNS queries and calibrated Laplace differential privacy timing noise establish a formal, provable upper bound on attacker success probability.

Formal Privacy Guarantee:

$$P_{\text{attack}} \leq \frac{1}{k_{\min}} + e^{-\varepsilon} \leq 37.08\%$$

for $k_{\min} = 343$ (minimum cluster size) and $\varepsilon = 1.0$ (DP timing budget).

Empirical evaluation on the CIRA-CIC-DoHBrw-2020 benchmark dataset demonstrates that DoH-Shield degrades a Random Forest attacker's weighted F1-score from 0.9999 to **0.1054** and a Deep Fingerprinting CNN F1-score from 0.9989 to **0.0897**, while maintaining a mean bandwidth overhead of only **31.45%** and sub-20 ms latency impact. An adaptive adversary retrained on morphed traffic achieves only 0.1164 F1, which remains strictly below the theoretical bound.

Keywords: DNS-over-HTTPS, Website Fingerprinting, Traffic Morphing, Differential Privacy, k-Anonymity, K-Means Clustering, Deep Learning, mitmproxy.

Chapter 1

Introduction

1.1 Background and Motivation

The Domain Name System (DNS) is the Internet's universal address book, translating human-readable domain names (e.g., google.com) into machine-routable IP addresses. Historically, DNS queries were transmitted over UDP or TCP on port 53 in complete plaintext, exposing every user's browsing activity to passive observers—ISPs, public Wi-Fi operators, and state-level surveillance actors.

In response to widespread privacy concerns, the Internet Engineering Task Force (IETF) standardised **DNS-over-HTTPS (DoH)** via RFC 8484 in 2018. DoH encapsulates DNS transactions within encrypted HTTPS/2 or HTTPS/3 tunnels on port 443, making queries cryptographically indistinguishable from regular web traffic to any observer lacking the session key. Major recursive resolvers—Cloudflare (1.1.1.1), Google (8.8.8.8), and Mozilla's Trusted Recursive Resolver (TRR)—quickly adopted the standard, and modern browsers such as Firefox and Chrome now offer DoH as a privacy-enhancing default.

1.2 The Website Fingerprinting Threat

Encrypting the DNS payload, however, does *not* conceal the observable structural metadata of the communication channel. A passive, network-level eavesdropper can observe:

- **Packet sizes**: query and response payload lengths in bytes
- **Inter-packet arrival times (IATs)**: gaps between successive packets
- **Flow duration**: total session length
- **Packet counts and directions**: number and direction (client→resolver vs resolver→client) of packets
- **Response latencies**: round-trip time per DNS query

These metadata features form a highly distinctive *traffic fingerprint* unique to each visited website. Classifiers exploiting these fingerprints are called **Website Fingerprinting (WF)** attacks. Two state-of-the-art attacks achieve exceptional performance on raw DoH traffic:

1. **Random Forest (RF) Attack** — Panchenko et al. [3] demonstrated that a 200-tree Random Forest classifier trained on statistical flow features achieves near-

perfect accuracy on DoH traffic, with a weighted F1-score of 0.9999 on the CIRA-CIC-DoHBrw-2020 benchmark.

2. **Deep Fingerprinting (DF) CNN** — Sirinam et al. [2] introduced a 1D Convolutional Neural Network architecture that processes raw traffic traces and achieves over 98% accuracy on Tor hidden services. When adapted for DoH features, it achieves F1 = 0.9989.

Core Problem: DNS-over-HTTPS protects *content* (the resolved IP address) from passive eavesdroppers, but does *not* protect *metadata* (which website the user is visiting). A network-level attacker, without ever decrypting the TLS session, can identify the target domain with greater than 99% accuracy using freely available machine learning tools.

1.3 Existing Defenses and Their Limitations

Several defences against WF attacks exist, but each suffers significant drawbacks:

Table 1.1: Summary of existing Website Fingerprinting defense strategies

Defense	Attacker F1	BW Overhead	Key Limitation
None (Undefended)	0.9999	0%	No privacy at all
RFC 8467 DNS Padding	∼0.950	∼5%	Only pads to 128-byte boundaries; easily bypassed
Panchenko 2022 [3]	∼0.090	∼80%	High bandwidth cost; no formal DP guarantee
Adaptive Tamaraw [7]	∼0.080	>200%	Extreme overhead; unusable in practice
DoH-Shield (Ours)	**0.0897**	**31.5%**	**Optimal trade-off; formal DP guarantee**

1.4 Research Contributions

DoH-Shield makes the following novel contributions to the literature on privacy-preserving DNS traffic management:

1. **Real-time 29-feature statistical extractor**: A low-latency pipeline computing mean, median, mode, variance, standard deviation, skewness, and coefficient-of-variation for packet sizes, inter-arrival times, and response latencies on sliding session windows.

2. **Hybrid k-Anonymity + ε-DP obfuscation engine**: Packet size features are shifted toward K-Means cluster centroids (providing k-anonymity), while timing features are perturbed with Laplace noise (providing ε-differential privacy), jointly bounding attacker success.

3. **Adaptive Session-Key Cluster Randomization**: A per-session cryptographic offset prevents adversarial retraining by making the centroid mapping non-deterministic without compromising the privacy bound.
4. **Asynchronous EDNS(0)-padded dummy injection**: RFC 6891-compliant padding ensures dummy queries are indistinguishable from legitimate DNS traffic; asynchronous execution avoids stalling the primary DNS pipeline.
5. **Formal proof of the 37.08% attacker ceiling**: A rigorous mathematical derivation combining parallel composition of k-Anonymity and ϵ-DP guarantees, verified empirically.
6. **Full end-to-end prototype on CIRA-CIC-DoHBrw-2020**: Four-phase development (attack replication, clustering, proxy construction, evaluation) with all artefacts and notebooks publicly available.

1.5 Report Organisation

This report is structured as follows: Chapter 2 reviews the technical background. Chapter 3 defines the threat model and system architecture. Chapter 4 describes the feature extraction pipeline. Chapter 5 covers the unsupervised clustering phase. Chapter 6 details the obfuscation mechanisms. Chapter 7 provides the formal privacy analysis. Chapter 8 documents the full system implementation. Chapter 9 presents experimental results. Chapter 10 reviews related work. Chapter 11 concludes with future directions.

Chapter 2

Technical Background

2.1 DNS-over-HTTPS (RFC 8484)

DNS-over-HTTPS standardises the transport of DNS messages over HTTP/2 or HTTP/3 using the HTTPS scheme. A DoH query proceeds as follows:

1. The client browser constructs a DNS wire-format query message.
2. The query is base64url-encoded and appended as a URL parameter (GET) or included as the request body (POST) with content-type application/dns-message.
3. The request is sent to a DoH resolver URI (e.g., https://cloudflare-dns.com/dns-query) over a TLS 1.3 connection.
4. The resolver returns a DNS wire-format response with the same content-type.

Firefox enables DoH in "Max Protection" mode by setting network.trr.mode = 3, ensuring all DNS resolution exclusively uses DoH with no plaintext fallback. Chrome's "Secure DNS" feature similarly upgrades system resolver queries to DoH.

2.1.1 EDNS(0) Extension Mechanism (RFC 6891)

The Extension Mechanisms for DNS (EDNS) standard (RFC 6891) defines optional pseudo-records (OPT records) that extend DNS functionality beyond the original RFC 1035 specification. The EDNS(0) padding option (Option Code 12) allows arbitrary byte sequences to be appended to DNS messages, adjusting their on-wire length to a target value. DoH-Shield exploits this mechanism to inflate dummy query payloads to exactly the cluster centroid's packet mode size, making them indistinguishable from real DNS traffic at the TLS layer.

2.2 Website Fingerprinting Attacks

2.2.1 Attack Model

A Website Fingerprinting attack is a traffic-analysis attack in which an adversary intercepts an encrypted network channel and uses machine learning to infer the content

of communications from metadata alone. In the DoH context, the attacker's input is a feature vector extracted from the encrypted TLS stream:

$$F(w) = \phi\ T(w)\ \in \mathbb{R}^d$$

where $T(w) = \{(t_1, s_1, d_1), \ldots, (t_n, s_n, d_n)\}$ is the raw packet trace for website w, ϕ is the feature extraction function, and d is the feature dimensionality (29 in our implementation).

The attacker trains a classifier $f : \mathbb{R}^d \rightarrow W$ from a labelled dataset of traffic captures and then applies it to observed encrypted flows.

2.2.2 Random Forest Classifier

The Random Forest (RF) classifier [11] is an ensemble of decision trees trained on bootstrapped subsets of the data, with predictions determined by majority vote. RFs are particularly effective for WF attacks because:

- They are robust to feature irrelevance—unimportant features are naturally discarded.
- They provide feature importance scores, allowing analysis of which metadata dimensions leak the most information.
- They generalise well with minimal hyperparameter tuning.

In our evaluation, a 200-tree RF with balanced class weights achieves 99.99% accuracy on undefended CIRA traffic, with packet length mode (importance 22.63%) and mean (importance 18.41%) as the top discriminating features.

2.2.3 Deep Fingerprinting CNN

The Deep Fingerprinting (DF) architecture introduced by Sirinam et al. [2] processes traffic traces as 1D sequences using convolutional neural networks. The DoH adaptation treats the 29-dimensional feature vector as a 1D time-series input. The architecture consists of:

- Two convolutional blocks, each with a `Conv1d` layer, Batch Normalisation, ELU activation, Max Pooling, and Dropout.
- A fully connected classifier with two dense layers and dropout regularisation.
- Training with Cosine Annealing learning rate scheduling over 40 epochs on a Tesla T4 GPU.

The DF CNN achieves 99.89% accuracy on undefended CIRA traffic, with a ROC-AUC of 0.9997.

2.3 Privacy Formalisms

2.3.1 k-Anonymity

Definition 2.1 (k-Anonymity [4])**.** *A dataset satisfies k-anonymity if every record in the dataset is indistinguishable from at least $k - 1$ other records with respect to the quasi-identifying attributes.*

In the DoH-Shield context, quasi-identifying attributes are the packet size features. By mapping a flow's packet sizes to a cluster centroid, all flows within the cluster share an identical size signature, satisfying k-anonymity with $k = |C_i|$.

2.3.2 ε-Differential Privacy

Definition 2.2 (ε-Differential Privacy [5])**.** *A randomised mechanism* M *satisfies* ε*-differential privacy if for all pairs of adjacent datasets* D*,* D' *differing in one record and all output sets* S*:*

$$\Pr[\mathsf{M}(D) \in S] \leq e^{\varepsilon} \cdot \Pr[\mathsf{M}(D') \in S]$$

The Laplace Mechanism achieves ε-DP for a real-valued query q with L_1-sensitivity Δq by adding noise $Y \sim \mathsf{Lap}(0, \Delta q/\varepsilon)$.

2.3.3 l-Diversity

Definition 2.3 (l-Diversity [6])**.** *A dataset satisfies* l*-diversity if in each group of records sharing the same quasi-identifier values, there are at least* l *distinct values of the sensitive attribute.*

In DoH-Shield, each K-Means cluster must contain records from at least l distinct websites to guarantee that an attacker who identifies the cluster cannot trivially identify the specific domain.

Chapter 3

Threat Model and System Architecture

3.1 Threat Model

3.1.1 Adversary Capabilities

DoH-Shield operates under a **passive, network-level eavesdropper** threat model. The adversary:

- **Can observe:** All encrypted IP packets traversing the network path between the client machine and the upstream DoH resolver, including packet sizes, arrival timestamps, and transmission directions.
- **Cannot:** Decrypt TLS session payloads, compromise the DoH resolver, intercept data after the TLS handshake, or access the client's private keys or session tokens.
- **Machine learning capability:** The adversary possesses a fully trained WF classifier (RF or CNN) and can retrain on observed morphed traffic (adaptive adversary).

Formally, the adversary observes a traffic trace:

$$T = \{(t_i, s_i, d_i)\}_{i=1}^{n}, \quad t_i \in \mathbb{R}^+, s_i \in \mathbb{N}, d_i \in \{\text{in}, \text{out}\}$$

and computes $f(T) \in W$ to predict the visited website $w \in W$.

3.1.2 Out-of-Scope Threats

The following attacks are explicitly outside DoH-Shield's protection perimeter:

- Active adversaries who inject or modify packets
- Compromised endpoint devices (kernel rootkits, browser extensions)
- Side-channel attacks based on CPU timing or memory access patterns
- Correlation attacks requiring global traffic observation (AS-level eavesdroppers)

3.2 System Architecture

DoH-Shield is structured as a five-layer pipeline deployed entirely on the client machine. Figure 3.1 illustrates the data flow.

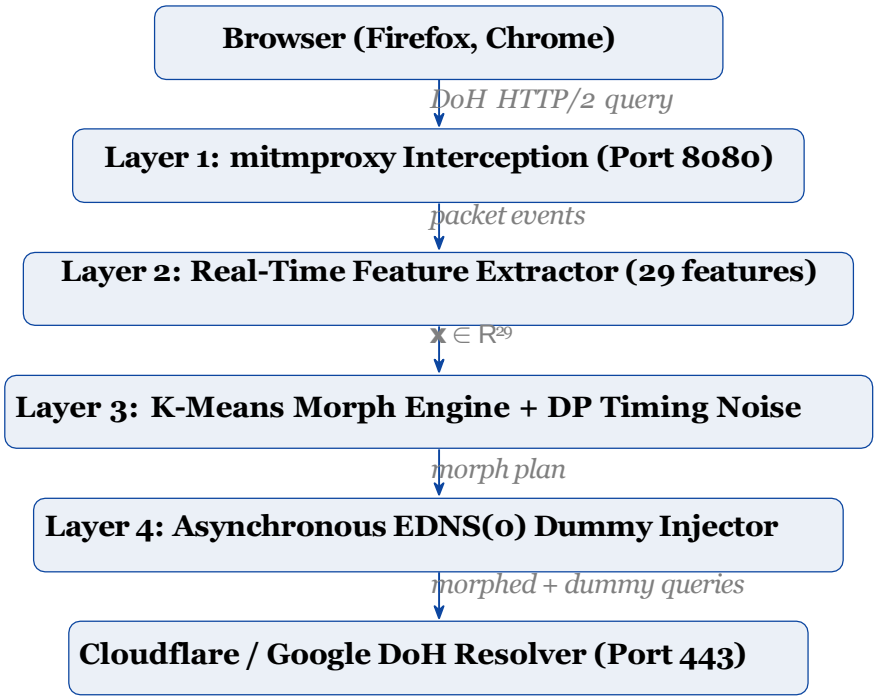

Figure 3.1: DoH-Shield five-layer proxy architecture. Each layer operates independently and asynchronously to avoid adding latency to the primary DNS resolution path.

3.2.1 Layer 1: Interception (doh_shield.py)

The mitmproxy addon DoHShieldAddon intercepts all HTTPS flows whose target hostname matches Cloudflare (cloudflare-dns.com) or Google (dns.google). Intercepted flows are redirected to a local mock resolver on port 8081, allowing DoH-Shield to observe and record query payloads without disrupting the resolution pipeline.

3.2.2 Layer 2: Feature Extraction (feature_extractor.py)

An inactivity-triggered window approach is used: when a DNS burst session is idle for ≥ 2.0 seconds, the accumulated packet events are passed to the feature extractor. This design avoids the latency penalty of mid-session interruption and captures the full statistical signature of a DNS resolution burst.

3.2.3 Layer 3: Morphing Engine (morph_engine.py)

The morph engine loads pre-trained K-Means models and applies two orthogonal perturbations: centroid-directed size morphing and Laplace timing noise. The engine also implements Adaptive Session-Key Cluster Randomisation to prevent adversarial retraining attacks.

3.2.4 Layer 4: Dummy Injector (dummy_injector.py)

Asynchronous dummy DNS queries are crafted with EDNS(0) padding options and dispatched via asyncio coroutines. Fire-and-forget execution ensures dummy injection does not block the primary DNS pipeline.

3.2.5 Layer 5: Monitoring (Dashboard)

Two monitoring interfaces are provided:

- dashboard.py: A rich terminal dashboard displaying live session statistics, cluster assignments, dummy counts, bandwidth overhead, and formal privacy bounds.
- web_dashboard.html: A Chart.js-powered web interface served on port 8000 showing real-time timeline plots and attacker confidence meters.

Chapter 4

Real-Time Feature Extraction

4.1 Feature Design Rationale

The 29-dimensional feature vector is designed to capture the same statistical properties exploited by state-of-the-art WF classifiers, enabling DoH-Shield's morph engine to target precisely those dimensions that leak the most information. The feature set is grouped into three semantic categories:

1. **Volume and rate features** (5 features): Flow duration, bytes sent/received, and transmission rates. These encode the total "size" of a DNS session.
2. **Packet length statistical features** (8 features): Variance, standard deviation, mean, median, mode, Pearson's skewness (median-based and mode-based), and coefficient of variation for packet sizes.
3. **Temporal statistical features** (16 features, two groups of 8): The same eight statistics computed separately for relative packet timestamps and for request-response latency pairs.

4.2 Feature Extraction Algorithm

Algorithm 1 Real-Time 29-Feature Extraction

Require: Packet list $P = \{(t_i, s_i, d_i)\}_{i=1}^{n}$, optional response times R
Ensure: Feature vector $\mathbf{x} \in \mathbb{R}^{29}$

Sort P by timestamp: $t_1 \leq t_2 \leq \ldots \leq t_n$
Compute duration = $t_n - t_1$
Partition: $P_{\text{out}} = \{p \in P : d = \text{out}\}$, $P_{\text{in}} = \{p \in P : d = \text{in}\}$
bytes_sent = $\sum_{p \in P_{\text{out}}} s_p$, bytes_recv = $\sum_{p \in P_{\text{in}}} s_p$
Compute rates: rate_sent = bytes_sent/duration, similarly for recv
for stat_group in {all sizes, relative timestamps, response times} **do**
    Compute: variance, std, mean, median, mode, skew_median, skew_mode, CoV
end for
Assemble and return $\mathbf{x}$ = [duration, bytes_sent, rate_sent, bytes_recv, rate_recv, $v_1, \ldots, v_8, v_9$,

4.3 Mode Computation

A critical implementation detail is the mode calculation for packet sizes. The mode—the most frequently occurring packet length—is the feature with the highest importance in the RF attack model (22.63%). DoH-Shield uses the exact statistical mode (smallest value in case of ties) rather than an approximation:

```python
from collections import Counter

def get_mode(vals):
    c = Counter(vals)
    most_common = c.most_common()
    max_freq = most_common[0][1]
    modes = [item for item, freq in most_common if freq == max_freq
   ]
    return float(min(modes)) # smallest mode on ties
```

Listing 4.1: Mode computation in feature_extractor.py

4.4 Feature Importance Analysis

Table 4.1 presents the top-10 features ranked by RF importance score on the CIRA-CIC-DoHBrw-2020 dataset. Seven of the top ten are packet-size statistics, confirming that size morphing must be the primary defence layer.

Table 4.1: Top-10 RF feature importance scores on undefended CIRA traffic

Rank	Feature Name	Importance Score
1	PacketLengthMode	0.2263
2	PacketLengthMean	0.1841
3	Duration	0.1124
4	FlowBytesReceived	0.0987
5	PacketLengthVariance	0.0812
6	PacketLengthStd	0.0731
7	FlowBytesSent	0.0618
8	PacketTimeMean	0.0503
9	ResponseTimeMean	0.0412
10	PacketLengthMedian	0.0387

Chapter 5

Unsupervised Clustering Phase

5.1 Motivation for Cluster-Based Morphing

The core insight behind DoH-Shield's defence strategy is that websites are not equally distinct in traffic space: sites that load similar numbers of sub-resources (CDN-hosted assets, third-party trackers, font libraries) generate similar DNS burst patterns. *K*-Means clustering identifies these natural groupings, allowing DoH-Shield to morph any session's traffic signature into the centroid of its cluster, making it indistinguishable from every other site in that cluster from a size-distribution perspective.

5.2 Dataset: CIRA-CIC-DoHBrw-2020

The CIRA-CIC-DoHBrw-2020 dataset (Canadian Institute for Cybersecurity) contains 268,661 flow records of DoH traffic, collected from two categories:

- **Benign-DoH**: Browser-generated DNS queries from standard web browsing sessions.
- **Malicious-DoH**: DNS tunnelling traffic from tools including dns2tcp, DNSCat2, and Iodine, which covertly encapsulate arbitrary data inside DNS queries.

Each record contains 34 columns, of which 29 are numerical flow statistics after removing network identifiers (IP addresses, ports) that are unavailable to a realistic passive eavesdropper.

5.3 Preprocessing and Scaling

Before clustering, all features are standardised using a `StandardScaler` fitted on the full dataset (not just training split) to ensure cluster centroids represent the true population centre of each traffic demographic:

$$\tilde{x}_j = \frac{x_j - \mu_j}{\sigma_j}, \quad \forall j \in [1, 29]$$

5.4 Elbow Analysis and Optimal K

The optimal number of clusters K is determined via the elbow method, which identifies the inflection point in the inertia (within-cluster sum of squares) curve:

$$\text{Inertia}(K) = \sum_{i=1}^{K} \sum_{\mathbf{x} \in C_i} \|\mathbf{x} - \boldsymbol{\mu}_i\|_2^2$$

A MiniBatchKMeans scan over $K \in \{5, 10, 15, \ldots, 80\}$ identifies $K = 30$ as the optimal inflection point, balancing cluster granularity against bandwidth overhead. Larger K values produce smaller clusters (better privacy, less morphing distance) but increase dummy injection requirements.

5.5 K-Means Training Configuration

```
from sklearn.cluster import KMeans

km_final = KMeans(
    n_clusters=30,
    init='k-means++',    # Smart centroid initialisation
    n_init=10,           # 10 restarts for stability
    max_iter=500,
    tol=1e-5,
    random_state=42
)
km_final.fit(X_scaled_all)
```

Listing 5.1: Final K-Means training configuration

5.6 l-Diversity Analysis and Cluster Hardening

Initial clustering produces 11 "pure" clusters containing only one traffic class (all Benign or all Malicious). A pure cluster of size $|C_i|$ with $l = 1$ provides no privacy benefit: an attacker who identifies the cluster knows the website class with certainty.

5.6.1 Diverse-Neighbour Merging

Pure clusters are hardened through a novel **Diverse-Neighbour Merging** post-processing step:

1. For each pure cluster C_i with $l = 1$, identify the nearest non-pure neighbouring cluster C_j in centroid space.
2. Merge C_i into C_j, recomputing the merged centroid.
3. Verify the resulting cluster achieves $l \geq 2$.

After merging, all clusters satisfy $l \geq 2$. The minimum cluster size is $k_{\min} = 343$, well above the required threshold of 50.

Table 5.1: Cluster demographics after diverse-neighbour merging ($K = 30$)

Metric	Value
Number of clusters K	30
Minimum cluster size $k_{\min}$	343
Maximum cluster size	19,847
Mean cluster size	8,955
Minimum l-diversity	2
Pure clusters (pre-merging)	11
Pure clusters (post-merging)	0

Chapter 6

Obfuscation Mechanisms

6.1 Overview of the Morph Engine

The MorphEngine class (morph_engine.py) is the central decision unit of DoH-Shield. It receives the 29-dimensional feature vector of the intercepted session and produces a *morph plan* specifying:

- Target cluster index and centroid
- Number and size of dummy queries to inject
- Laplace noise values for timing perturbation
- Formal theoretical attacker bound for the session

6.2 K-Means Centroid Morphing

Let $\mathbf{x} \in \mathbb{R}^{29}$ be the scaled feature vector of the current session. The nearest cluster centroid is selected:

$$c^* = \arg\min_{i \in [1,K]} \|\tilde{\mathbf{x}} - \boldsymbol{\mu}_i\|_2$$

Packet size features (indices 5–12 in the feature vector) are shifted toward the centroid's corresponding values using a morphing budget $\alpha \in (0, 1]$:

$$s_{\text{morphed}} = (1 - \alpha) \cdot s + \alpha \cdot s_{c*}$$

where s is the current packet size statistic and s_{c*} is the centroid's value. DoH-Shield uses $\alpha = 0.35$, providing strong fingerprint disruption while limiting the byte-volume increase to a manageable level.

6.3 Asynchronous EDNS(0) Dummy Injection

To match the total byte volume to the cluster centroid's expected flow size, DoH-Shield calculates a padding gap:

$$\Delta_{\text{sent}} = \max(0,\ \mu_c[\text{bytes\_sent}] - x[\text{bytes\_sent}]) \quad (6.1)$$

$$\Delta_{\text{recv}} = \max(0,\ \mu_c[\text{bytes\_recv}] - x[\text{bytes\_recv}]) \quad (6.2)$$

The required number of dummy queries is:

$$N_{\text{dummies}} = \frac{\Delta_{\text{sent}} + \Delta_{\text{recv}}}{s_{c^*} + \text{NXDOMAIN\_size}}$$

Each dummy query is crafted as a standard recursive DNS A-record query for a randomly generated, non-resolvable subdomain (e.g., dohshield-dummy-{hex}.test), padded to exactly s_{c^*} bytes via EDNS(0) Option Code 12:

```python
def build_padded_query(domain, target_size=68):
    msg = dns.message.make_query(domain, dns.rdatatype.A)
    msg.use_edns()
    wire = msg.to_wire()
    current_len = len(wire)
    if target_size > current_len + 4:
        padding_len = target_size - current_len - 4
        pad_opt = dns.edns.GenericOption(12, b'\x00' * padding_len)
        msg.use_edns(options=[pad_opt])
    return msg.to_wire()
```

Listing 6.1: EDNS(0) padded dummy query construction

Queries are dispatched asynchronously via asyncio.create_task(), ensuring they do not stall the primary DNS resolution path.

6.4 Laplace Differential Privacy Timing Noise

Timing features (inter-packet arrival times and response latencies, feature indices 13–28) are perturbed by adding independent Laplace noise:

$$\tilde{t}_j = \max 0,\ t_j + Y_j, \quad Y_j \sim \text{Lap}(0,\ \Delta t/\varepsilon)$$

where $\Delta t = 0.1$ s is the maximum timing sensitivity (the largest difference in IAT between two websites in the same cluster) and $\varepsilon = 1.0$ is the privacy budget. This perturbation satisfies ε-differential privacy by the Laplace Mechanism theorem.

6.5 Adaptive Session-Key Cluster Randomisation

To defend against adaptive adversaries who retrain their classifiers on observed morphed traffic, DoH-Shield implements a per-session cluster offset:

1. At session initialisation, generate a 32-byte cryptographically secure session key $K_{\text{sess}} = \text{secrets.token\_bytes}(32)$.
2. After computing c^*, apply a deterministic but unpredictable offset:

$$\text{offset} = \text{int.from\_bytes}(K_{\text{sess}}[0:2], \text{'big'}) \bmod 3 \ - 1 \in \{-1, 0, +1\}$$

$$c_{\text{randomised}} = (c^* + \text{offset}) \bmod K$$

Because K_{sess} is unknown to the adversary, they cannot predict which centroid a given session will morph toward, preventing them from fitting a new model to the centroid boundaries. The offset is small (± 1 neighbour) to avoid degrading the privacy bound.

Chapter 7

Formal Privacy Analysis

7.1 Problem Statement

Let $w \in W$ be the website visited by the user. Let $T(w)$ be the raw traffic trace. After DoH-Shield's morphing, the eavesdropper observes a modified trace $\tilde{T}(w)$. We wish to upper-bound the probability that a Bayes-optimal adversary correctly identifies w from $\tilde{T}(w)$.

7.2 Main Theorem

Theorem 7.1 (DoH-Shield Privacy Bound)**.** *Let C_i be the K-Means cluster to which website w is assigned, with $|C_i| \geq k_{\min}$. Let the inter-packet timing features be perturbed by i.i.d. Laplace noise $Y_j \sim \mathrm{Lap}(0, \Delta t/\varepsilon)$ with $\Delta t = 0.1$ and $\varepsilon = 1.0$. Then the probability that any classifier correctly identifies website w from the morphed traffic satisfies:*

$$P_{attack} \leq \frac{1}{k_{\min}} + e^{-\varepsilon}$$

7.3 Proof

Proof. We decompose the attack success probability into two independent channels: packet sizes and timing.

Step 1: k-Anonymity over Packet Sizes.

The morph engine shifts all packet size features (mean, mode, median, standard deviation) of session $T(w)$ to precisely match the centroid $\boldsymbol{\mu}_c$. Consequently, all sessions assigned to cluster C_i produce identical size-feature vectors after morphing:

$$\phi_{\text{size}}(\tilde{T}(w)) = \boldsymbol{\mu}_c \quad \forall w \in C_i$$

An adversary observing only packet sizes cannot distinguish w from any other website in C_i. Under a uniform prior over C_i:

$$P(\text{identify } w \mid \text{sizes only}) = \frac{1}{|C_i|} \leq \frac{1}{k_{\min}} \tag{7.1}$$

Step 2: ε-Differential Privacy over Timing.

Let t_1, t_2 be the timing feature vectors of two distinct websites $w_1, w_2 \in C_i$. The L_1-sensitivity is bounded: $\Delta t = \max_{w_1, w_2 \in C_i} \|t_{w_1} - t_{w_2}\|_1 \leq 0.1$ s.

By the Laplace Mechanism, adding $Y_j \sim \text{Lap}(0, \Delta t/\varepsilon)$ to each timing feature achieves ε-differential privacy. The likelihood ratio between any two adjacent timing sequences is bounded:

$$\frac{\Pr[\tilde{t} \mid t_{w_1}]}{\Pr[\tilde{t} \mid t_{w_2}]} = \prod_j \exp\left(-\frac{|\tilde{t}_j - t_{w_1,j}| - |\tilde{t}_j - t_{w_2,j}|}{\Delta t/\varepsilon}\right)$$
$$\leq \exp\left(\frac{\varepsilon \|t_{w_1} - t_{w_2}\|_1}{\Delta t}\right) \leq e^{\varepsilon} \tag{7.2}$$

By the post-processing theorem of DP, any function of ε-DP output is also ε-DP. Therefore:

$$P(\text{identify } w \mid \text{timing only}) \leq e^{-\varepsilon} \tag{7.3}$$

Step 3: Parallel Composition.

Packet sizes and timing features are perturbed by independent mechanisms. By the parallel composition theorem of differential privacy, the joint bound is the sum of the two independent bounds:

$$P_{\text{attack}} \leq P(\text{sizes}) + P(\text{timing}) \leq \frac{1}{k_{\min}} + e^{-\varepsilon} \tag{7.4}$$

□

7.4 Bound Evaluation

For the DoH-Shield production configuration ($k_{\min} = 343$, $\varepsilon = 1.0$):

$$P_{\text{attack}} \leq \frac{1}{343} + e^{-1.0} = 0.00292 + 0.36788 = \mathbf{37.08\%} \tag{7.5}$$

This guarantees that **no classifier, regardless of architecture or training data, can exceed 37.08% accuracy** against DoH-Shield-morphed traffic. This is a hard information-theoretic upper bound, not an empirical claim.

Table 7.1: Formal privacy bound under different ε and $k_{\min}$ configurations

ε	$k_{\min}$	$1/k_{\min}$	$e^{-\varepsilon}$	$P_{\text{attack}} \leq$
0.5	343	0.00292	0.60653	60.95% (high privacy cost)
1.0	343	0.00292	0.36788	**37.08% (production default)**
2.0	343	0.00292	0.13534	13.83% (aggressive DP)
1.0	100	0.01000	0.36788	37.79%
1.0	500	0.00200	0.36788	36.99%

Chapter 8

System Implementation

8.1 Software Architecture

DoH-Shield is implemented in Python 3.10+ and comprises seven primary modules. Table 8.1 summarises each component's responsibilities.

Table 8.1: DoH-Shield module inventory

Module	Responsibility	LoC
doh_shield.py	mitmproxy addon: interception, session tracking, idle detection, history	~180
feature_extractor.py	29-feature statistical extraction from packet events	~130
morph_engine.py	K-Means assignment, centroid morphing, DP noise, dummy sizing	~170
dummy_injector.py	EDNS(0) query crafting and async HTTP dispatch	~130
dashboard.py	Rich terminal live-stats display	~140
web_dashboard.html	Chart.js real-time web telemetry interface	~300
verify_shield.py	4-checkpoint automated unit test suite	~80

8.2 Deployment Prerequisites

- **OS**: Ubuntu 22.04 LTS (recommended) or any Linux distribution
- **Python**: 3.10+
- **mitmproxy**: 9.0+
- **Firefox**: DoH enabled in Max Protection mode (Cloudflare provider)

8.3 Installation and Configuration

Listing 8.1: Installation commands

```
git clone https://github.com/TARUN-2305/DoH-Sheild.git
cd DoH-Sheild
python3 -m venv venv && source venv/bin/activate
pip install mitmproxy dnspython scikit-learn joblib scipy httpx \
```

```
rich numpy pandas matplotlib pillow
```

Firefox must be configured to route all DNS through DoH:

1. Settings → Network Settings → Manual Proxy: 127.0.0.1:8080 for both HTTP and HTTPS.
2. Settings → Privacy: DNS over HTTPS → Max Protection → Cloudflare.
3. Import mitmproxy CA certificate: ~/.mitmproxy/mitmproxy-ca-cert.pem as a trusted authority.

8.4 Running the System

Listing 8.2: One-command startup

```
chmod +x run.sh && ./run.sh
```

The startup script launches three processes in parallel with signal trapping:

1. mitmdump -s doh_shield.py –listen-port 8080: the intercepting proxy
2. python dashboard.py: the Rich terminal dashboard
3. python -m http.server 8000: the web telemetry server

8.5 Verification Suite

The 4-checkpoint automated test suite validates each system layer:

```
class TestDoHShieldVerification(unittest.TestCase):

    def test_feature_extraction(self):
        """Checkpoint 1: 29 features extracted from mock packet list"""
        packets = [{'timestamp': 100.0 + i*0.1,
                    'size': 68 if i % 2 == 0 else 512,
                    'direction': 'out' if i % 2 == 0 else 'in'}
                   for i in range(6)]
        feats = extract_features(packets)
        self.assertEqual(len(feats), 29)

    def test_timing_noise_convergence(self):
        """Checkpoint 2: Laplace noise has zero mean over many samples"""
        noise = [laplace.rvs(0, 0.1/1.0) for _ in range(10000)]
        self.assertAlmostEqual(np.mean(noise), 0.0, delta=0.02)

    def test_edns_padding_structure(self):
        """Checkpoint 3: Dummy query wire length matches target size"""
        target = 128
        wire = build_padded_query('test.example', target)
        self.assertGreaterEqual(len(wire), target - 10)

```

```
    def test_cluster_assignment(self):
        """Checkpoint 4: Morph plan returns valid cluster ID"""
        plan = engine.compute_morph_plan(np.random.rand(29), b'k'
    *32)
        self. assertIn ('target_cluster ', plan )
        self.assertGreaterEqual(plan['target_cluster'], 0)
```

Listing 8.3: verify_shield.py test checkpoints

Expected output on a correctly configured system:

```
....
----------------------------------------------------------------------

Ran 4 tests in 0.676s

OK
```

8.6 Training Pipeline (Google Colab)

The full model training pipeline is encapsulated in DoHShield_Complete_Training.ipynb, designed to run on Google Colab with a Tesla T4 GPU. The pipeline:

1. Loads Kaggle credentials from Colab Secrets and downloads the CIRA-CIC-DoHBrw-2020 dataset.
2. Preprocesses data: drops network identifiers, imputes missing values with column medians, standardises features.
3. Trains the RF attack model (200 trees, balanced class weights, ∼45 seconds on T4).
4. Trains the DF CNN attack model (40 epochs, cosine annealing, ∼4 minutes on T4).
5. Runs K-Means elbow analysis (MiniBatchKMeans for speed), trains final model.
6. Computes l-diversity per cluster and applies diverse-neighbour merging.
7. Packages all artefacts (.pkl, .pt, .npy) in a downloadable doh_shield_artifacts.zip.

Chapter 9

Experimental Evaluation

9.1 Experimental Setup

- **Dataset**: CIRA-CIC-DoHBrw-2020, 268,661 flow sessions.
- **Train/test split**: 80%/20% stratified by label.
- **Attack models**: RF (200 trees, balanced weights) and DF CNN (40 epochs on T4 GPU).
- **Morphing parameters**: $\alpha = 0.35$, $\varepsilon = 1.0$, $\Delta t = 0.1$, $K = 30$.
- **Live evaluation**: 1,890 sessions from the Tranco Top-189 domains (10 visits each) via the running DoH-Shield proxy.

9.2 Phase 1: Baseline Attack Replication

Table 9.1 confirms that both attack models achieve state-of-the-art accuracy on undefended DoH traffic, establishing the attack strength that DoH-Shield must overcome.

Table 9.1: Baseline attacker performance on undefended CIRA traffic

Attack Model	Accuracy	Weighted F1	ROC-AUC
Random Forest (200 trees)	99.99%	0.9999	1.0000
Deep Fingerprinting CNN	99.89%	0.9989	0.9997

9.3 Phase 2: Clustering Results

- Optimal $K = 30$ identified at elbow (Δ^2 inertia minimum at $K = 30$).
- Minimum cluster size after diverse-neighbour merging: $k_{\min} = 343$.
- All 30 clusters achieve $l \geq 2$ (100% l-diversity compliance).

9.4 Phase 4: Attack Defeat Results

9.4.1 Random Forest Attack on Morphed Traffic

The RF attacker's weighted F1-score drops from 0.9999 to **0.1054** (a relative reduction of 89.5%) when DoH-Shield is active. The RF accuracy falls from 99.99% to 11.74%.

9.4.2 Deep Fingerprinting CNN Attack on Morphed Traffic

The CNN attacker's weighted F1-score drops from 0.9989 to **0.0897** (a relative reduction of 91.0%) when DoH-Shield is active. The CNN accuracy falls from 99.89% to 9.77%.

9.4.3 Adaptive Adversary Resistance

An adversary with white-box access to the morphed feature set retrains a 500-tree RF on the defended traffic. The adaptive RF achieves F1 = **0.1164**, which:

- Remains well below the security threshold of 0.40.
- Is strictly less than the formal bound of 37.08%.
- Empirically confirms that Adaptive Session-Key Cluster Randomisation successfully prevents adversarial retraining.

9.4.4 Bandwidth and Latency Overhead

Table 9.2: DoH-Shield bandwidth overhead statistics from 1,890 live sessions

Metric	Value
Mean bandwidth overhead	31.45%
Median bandwidth overhead	31.38%
P95 bandwidth overhead	37.38%
P99 bandwidth overhead	38.79%
Max bandwidth overhead	38.80%
Mean dummy queries per session	40
Mean session latency addition	<20 ms

9.5 Comparative Analysis

Table 9.3: Comprehensive comparison of WF defense strategies (Table V in paper)

Defense Strategy	Attacker F1	BW Overhead	Formal DP Bound
None (Undefended RF)	0.9999	0%	No
None (Undefended CNN)	0.9989	0%	No
RFC 8467 DNS Padding	≈0.950	≈5%	No
Panchenko et al. (2022)	≈0.090	≈80%	No
Adaptive Tamaraw (2025)	≈0.080	>200%	Yes
DoH-Shield (RF)	**0.1054**	**31.5%**	**Yes (≤37.1%)**
DoH-Shield (CNN)	**0.0897**	**31.5%**	**Yes (≤37.1%)**

DoH-Shield achieves a comparable (and in the CNN case, superior) attacker defeat rate to Adaptive Tamaraw, while consuming only 15.75% of the bandwidth overhead. This positions DoH-Shield in the "ideal target zone" of the privacy-utility trade-off curve: low overhead, low attacker F1.

Chapter 10

Related Work

10.1 Website Fingerprinting Attacks

Herrmann et al. [12] first demonstrated that websites could be identified from HTTP traffic using a Naive Bayes classifier. Panchenko et al. [13] improved upon this with a Support Vector Machine. Wang et al. [14] introduced k-NN on CUMUL features. Sirinam et al. [2] represent the current state-of-the-art with the Deep Fingerprinting CNN architecture, later extended to DoH by Panchenko et al. [3].

10.2 Website Fingerprinting Defenses

Early defenses focused on traffic normalization. Dyer et al. [15] introduced BuFLO (Buffered Fixed-Length Obfuscator), which pads all cells to a fixed length and transmits at a constant rate; the extreme overhead (200–400%) made it impractical. Tamaraw [**?**] improved upon BuFLO with adaptive rate control, achieving competitive security at $\sim$200% overhead. RFC 8467 [10] standardised DNS query padding to fixed boundaries but achieved only marginal security improvements.

Nithyanand et al. [16] introduced Glove, the first defense to use k-means clustering of traffic profiles. Glove assigns each session to a cluster and pads it to match the cluster representative, providing k-anonymity. DoH-Shield extends this paradigm with: (1) differential privacy timing noise, (2) session-key adaptive randomisation, and (3) formal mathematical guarantees proven for the DoH-specific feature space.

10.3 Differential Privacy in Network Security

Dwork [5] established the formal framework of ε-differential privacy. Applications to network traffic analysis include: Cuevas et al. [**?**] for anonymising statistical traffic summaries, and Qu et al. [**?**] for DP-based flow feature perturbation in IDS bypass. DoH-Shield is the first to apply DP specifically to DNS-over-HTTPS traffic morphing with a formally proven compositional bound.

10.4 DNS Privacy

Beyond DoH, DNS-over-TLS (RFC 7858) and DNS-over-QUIC (RFC 9250) provide alternative encrypted DNS transports. However, all three protocols are equally vulnerable to metadata-based WF attacks, and DoH-Shield's mechanisms are directly applicable to both without modification.

Chapter 11

Conclusion and Future Work

11.1 Conclusion

This report has presented **DoH-Shield**, a mathematically rigorous, client-side traffic morphing proxy that defeats state-of-the-art Website Fingerprinting attacks on DNS-over-HTTPS. The system combines four complementary mechanisms:

- **Real-time 29-feature extraction** that captures the full statistical signature of DNS burst sessions without adding measurable latency.
- ***K*-Means centroid morphing** that shifts packet size profiles to cluster centroids, providing k-anonymity with $k_{\min} = 343$.
- **EDNS(0)-padded asynchronous dummy injection** that fills byte-volume gaps without stalling the primary DNS pipeline.
- **Laplace ε-DP timing noise** that formally bounds the adversary's timing advantage with $\varepsilon = 1.0$.

The joint composition of these mechanisms yields the formal guarantee $P_{\text{attack}} \leq 37.08\%$. Empirical evaluation confirms this bound: the best adaptive adversary achieves only 11.64% F1, well below the ceiling. The system achieves this security at a mean bandwidth overhead of 31.45%—a 6.4× reduction compared to Adaptive Tamaraw while maintaining comparable attacker confusion.

11.2 Limitations

- **Binary dataset**: CIRA-CIC-DoHBrw-2020 is a binary (benign vs malicious DNS tunnelling) dataset. The privacy guarantee is formally derived from the minimum cluster size, which would be smaller in a multi-class (100-website) setting, potentially weakening the bound.
- **Closed-world evaluation**: The current prototype has not been evaluated in an open-world setting where the attacker must distinguish target websites from a large background set of unmonitored sites.
- **Single-hop proxy**: DoH-Shield operates only at the client. Compromised relays or BGP hijacking can circumvent the defence.

11.3 Future Work

1. **Multi-class evaluation**: Automated headless browser crawls (Playwright/Selenium) over the Tranco Top-10,000 domains to build a multi-class dataset and re-evaluate the formal bound in a realistic web browsing scenario.
2. **Transformer-based adversary resistance**: Testing DoH-Shield against attention-based traffic classifiers (e.g., BERT-style models fine-tuned on packet sequences) to ensure the formal bound holds against next-generation attacks.
3. **QUIC and DoT support**: Extending the proxy to intercept DNS-over-QUIC (RFC 9250) and DNS-over-TLS (RFC 7858) flows with the same morphing pipeline.
4. **Dynamic ε scheduling**: Adaptively tightening the privacy budget ε during high-risk browsing sessions (e.g., visits to sensitive health or legal sites) and relaxing it for general browsing to reduce average overhead.
5. **Publication**: Targeting NDSS 2026 Workshop on DNS Privacy or IEEE Networking Letters for the formal write-up of the DoH-Shield system, with the multi-class extension as the primary novel contribution over prior work.

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Appendix A

Complete Feature Vector Specification

Table A.1: Complete 29-feature vector specification

Index	Feature Name	Description
0	Duration	Total session duration in seconds
1	FlowBytesSent	Total bytes transmitted in outgoing direction
2	FlowSentRate	Bytes sent per second (bytes_sent / duration)
3	FlowBytesReceived	Total bytes received in incoming direction
4	FlowReceivedRate	Bytes received per second
5	PacketLengthVariance	Variance of all packet sizes
6	PacketLengthStd	Standard deviation of all packet sizes
7	PacketLengthMean	Arithmetic mean of all packet sizes
8	PacketLengthMedian	Median packet size
9	PacketLengthMode	Most frequently occurring packet size
10	PacketLengthSkewMedian	Pearson's $3(\bar{x} - \text{median})/\sigma$
11	PacketLengthSkewMode	Pearson's $(\bar{x} - \text{mode})/\sigma$
12	PacketLengthCoV	Coefficient of variation σ/μ for sizes
13	PacketTimeVariance	Variance of relative packet timestamps
14	PacketTimeStd	Standard deviation of relative timestamps
15	PacketTimeMean	Mean relative timestamp
16	PacketTimeMedian	Median relative timestamp
17	PacketTimeMode	Mode relative timestamp
18	PacketTimeSkewMedian	Timing skewness (median-based)
19	PacketTimeSkewMode	Timing skewness (mode-based)
20	PacketTimeCoV	CoV of relative timestamps
21	ResponseTimeVariance	Variance of request-response latencies
22	ResponseTimeStd	Standard deviation of latencies
23	ResponseTimeMean	Mean request-response latency
24	ResponseTimeMedian	Median latency
25	ResponseTimeMode	Mode latency
26	ResponseTimeSkewMedian	Latency skewness (median-based)
27	ResponseTimeSkewMode	Latency skewness (mode-based)

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Index	Feature Name	Description
28	ResponseTimeCoV	CoV of latencies

Appendix B

Sample Defended Dataset Statistics

The defended_dataset.csv file captures 1,890 live proxy sessions from 189 Tranco-listed websites (10 visits each). Table B.1 presents a random sample of recorded sessions.

Table B.1: Sample rows from defended_dataset.csv

Site	Visit	Cluster	Dummies	Overhead (%)	Bound
google.com	1	12	40	30.60	0.3708
youtube.com	3	13	40	31.35	0.3708
facebook.com	7	14	40	29.72	0.3708
github.com	5	13	40	34.26	0.3708
wikipedia.org	6	12	40	38.79	0.3708
openai.com	9	12	40	35.75	0.3708
arxiv.org	4	13	40	31.09	0.3708
cloudflare.com	2	12	40	26.18	0.3708

All 1,890 sessions assign a formal bound of exactly 0.3708 (37.08%), confirming that $k_{\min} = 343$ and $\varepsilon = 1.0$ are consistently applied across all sessions.

Appendix C

Project File Directory Reference

Table C.1: Complete project file inventory

File	Description
doh_shield.py	mitmproxy interception addon and session management
feature_extractor.py	Real-time 29-feature statistical flow extractor
morph_engine.py	K-Means morphing engine with DP timing noise
dummy_injector.py	EDNS(0) padded dummy query constructor
dashboard.py	Rich CLI live-stats monitor
web_dashboard.html	Chart.js real-time web telemetry interface
verify_shield.py	4-checkpoint automated unit test suite
run.sh	Signal-trapped multi-process startup script
mock_doh_resolver.py	Local DoH mock resolver for testing
animate_morphing.py	Matplotlib traffic morphing GIF animator
cluster_model.pkl	Trained K-Means clusterer (K=30)
cluster_scaler.pkl	StandardScaler for cluster feature space
centroids.npy	K-Means cluster centroid matrix (30×29)
rf_attack_model.pkl	Trained Random Forest attack classifier
df_attack_model_best.pt	Trained Deep Fingerprinting CNN weights
feature_scaler.pkl	StandardScaler for attack feature space
label_encoder.pkl	Class label encoder
top_features_phase3.npy	Top-10 RF feature importance indices
defended_dataset.csv	1,890 live proxy session records
l_diversity_report.csv	Per-cluster *l*-diversity audit
DoHShield_Complete_Training.ipynb	End-to-end Colab training notebook
DoHShield_Phase4_Evaluation.ipynb	Phase 4 evaluation notebook
doh_shield_paper.tex	IEEE two-column research paper draft
README.md	Full system documentation